

USHA AAVULA

9408437683 ushareddyaaavula@gmail.com [LinkedIn](#)

PROFESSIONAL SUMMARY

I am an analytical and detail-oriented Data Science graduate from the University of North Texas with hands-on experience in machine learning, predictive analytics, and data visualization. Established ability to provide actionable conclusions from real-world data through school and personal projects. Keen to solve complex issues by leveraging data and seeking full-time opportunity to collaborate with a data-driven team.

ACADEMICS:

Master of Science in Data science	08/2023 – 05/2025
University of North Texas, Denton, Texas, USA	3.8 GPA
Bachelor of Technology in Electrical and Electronics Engineering	06/2019 – 06/2023
MallaReddy College of Engineering and Technology, India	

MS Coursework:

Data Science Tools: Python, R, SQL, Excel, Jupyter Notebooks, Google Colab, Git, GitHub

Data Visualization: Tableau, Power BI, Matplotlib, Seaborn, Visual Storytelling, KPI Dashboards

Machine Learning and Predictive Modeling: Supervised and Unsupervised Learning, Classification (Logistic Regression, Decision Trees, Random Forest), Regression (Linear Regression, Ridge, Lasso), Clustering (K-Means, Hierarchical), Neural Networks, Deep Learning (TensorFlow, PyTorch), Time Series Forecasting, Feature Engineering, Model Evaluation (Cross-Validation, GridSearchCV), A/B Testing, Model Deployment

Statistical Analysis: Descriptive and Inferential Statistics, Hypothesis Testing, ANOVA, Chi-Square Tests, Correlation and Covariance, Probability Distributions, Linear Programming, Forecasting Models (ARIMA, Exponential Smoothing)

Database Management: Relational Databases (MySQL, PostgreSQL), NoSQL (MongoDB), SQL (Joins, Subqueries, CTEs, Window Functions), ETL Pipelines, Data Warehousing (Snowflake, Redshift), Data Cleaning and Transformation

Big Data and Cloud Computing: Google Cloud Platform (BigQuery, Cloud Storage, Vertex AI), AWS (S3, Lambda, EC2), Databricks, Apache Spark, Hadoop Basics, Distributed Data Processing, Scalable Cloud-based Model Deployment

Projects:

Credit Card Customer Segmentation Using Clustering

- Applied K-Means clustering to segment credit card customers based on demographic and transactional data
- Developed machine learning models for predictive segmentation and targeted marketing
- Enhanced customer satisfaction and optimized marketing strategies
- Automated customer segmentation for future marketing campaigns

Diabetic Patients Record Analysis

- Used logistic regression and Random Forest models to predict hospital readmission risk
- Identified key factors (age, hospital stay length, diagnosis count) influencing readmission
- Recommended targeted interventions to reduce readmission rates

Employee Payroll Management System

- Designed a payroll management system using MySQL Workbench for data storage and management
- Streamlined payroll processing and improved data accuracy through structured workflows
- Secured sensitive employee data and ensured error-free payroll calculations

Comparative Sentiment Analysis of Twitter vs X — Data Analyst

Jan 2025

- Led a large-scale NLP project analyzing 300,000+ tweets across pre- and post-acquisition periods using Python and transformer-based models (Roberta, VADER) to evaluate sentiment shifts in user discourse.
- Developed a keyword-driven classification pipeline to extract moderation-related tweets, supporting sentiment comparison and policy impact assessment.
- Applied BERTopic for unsupervised topic modeling, revealing thematic evolution from emotional discourse (Twitter) to ideological and tech-centric discussion (X).

Comparative Analysis of Respiratory Diseases (Asthma vs. Influenza) — Data Analyst

- Analyzed MEPS (Medical Expenditure Panel Survey) data to compare asthma and influenza trends across demographics, healthcare costs, and medical visit frequency.
- Applied statistical methods including Chi-Square and Mann-Whitney U tests to uncover significant disparities between patient groups.
- Identified key factors impacting healthcare access and cost, informing strategies for equitable resource distribution.
- Presented findings through dashboards and data visualizations, delivering actionable insights to stakeholders in public health planning.
- Conducted Chi-Square hypothesis testing to validate sentiment distribution significance, uncovering that negative sentiment did not statistically increase post-transition.

Flood Analysis & Management System — Data Engineer / Analyst

- Developed a cloud-based flood risk analysis system using Google Cloud Platform (GCP), BigQuery, Hive, and Apache Spark for large-scale data processing and analysis.
- Cleaned and enriched geospatial and historical flood data using Python and OpenRefine to enable accurate querying and analytics.
- Designed interactive dashboards in Looker Studio to visualize damage patterns, weather trends, and fatal event distributions.
- Provided data-driven insights to support disaster preparedness for nonprofit organizations and laid the groundwork for real-time alerts and predictive modeling.

CERTIFICATIONS & ACHIEVEMENTS:

- Wipro Talent Next JAVA course completion certificate by Wipro

EXTRA CURRICULAR ACTIVITIES:

- Been a Class representative and led 60 students and collaborated with professors for annual meets and events.